Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2021/2022

COURSE NAME	:	DISCRETE STRUCTURE
COURSE CODE	:	BIC 10103
PROGRAMME CODE	:	BIS / BIP / BIW / BIM
EXAMINATION DATE	:	JANUARY / FEBRUARY 2022
DURATION	:	3 HOURS
INSTRUCTION	:	1. ANSWER **ALL** QUESTIONS. 2. THIS FINAL EXAMINATION IS CONDUCTED ONLINE AND OPEN BOOK.

THIS QUESTION PAPER CONSISTS OF **FOUR** (4) PAGES

TERBUKA

Q1 (a) Suppose that a function, f is defined recursively by

$$f(0) = 3$$
$$f(n+1) = 2f(n) + 3$$

Find the first five terms of the function.

(5 marks)

(b) Use mathematical induction to prove that

$$1^2 - 2^2 + 3^2 - .... + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$$

for all positive integers, n.

(15 marks)

Q2 (a) Let x and y be sets. Discuss the notation of $x\ R\ y$.

(2 marks)

(b) Identify **TWO (2)** ways to represent a relation.

(2 marks)

(c) Let R be a relation on $\{1, 2, 3, 4, 5\}$ defined by as follow:

$$R = \{(x, y) | \ x > y \wedge x + y \ is \ odd \ and \ prime\}$$

(i) Write down the domain and range of R.

(12 marks)

(ii) Sketch the graph of R.

(4 marks)

TERBUKA

Q3 (a) A male usurer lends money at unreasonable rates of interest. He demands to be paid 10% interest on a loan, compounded weekly. Suppose you borrow RM500 from him.

Calculate the total of money you owe if you wait two months to pay him back.

(5 marks)

(b) Formulate a solution to the following recurrence relations.

(i) $a_n = 3a_{n-1} - 2a_{n-2}$ where $a_0 = 1$ and $a_1 = 2$.

(8 marks)

(ii) $a_n = 6a_{n-1} - 9a_{n-2}$ where $a_0 = 1$ and $a_1 = 6$.

(7 marks)

Q4 (a) Estimate the time complexity, *T(n)* and Big-O notation, *O(n)* of the algorithm in **Figure Q4(a)**.

```
i = 1;
sum = 0;
while (i <= n)
{
      j = 1;
      while (j <= n)
      {
      sum = sum + 1;
      j = j + 1;
      }
      i = i + 1;
}
```

Figure Q4(a)

(10 marks)

(b) Write a pseudocode of an algorithm that calculates the sum of all integers in a list.

(5 marks)

(c) Estimate the time complexity, *T(n)* of the algorithm in **Q4(b)**.

(5 marks)

TERBUKA

Q5 (a) Five computers are connected in a network, and the given adjacency matrix indicates the cable connections between those computers.

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

(i) Draw a graph to illustrate the network configuration.

(5 marks)

(ii) Based on the graph in **Q5(a)(i)**, analyze whether the network is a Eulerian or Hamiltonian.

(5 marks)

(b) **Figure Q5(b)** shows some major roads in New Jersey with their distances between cities.

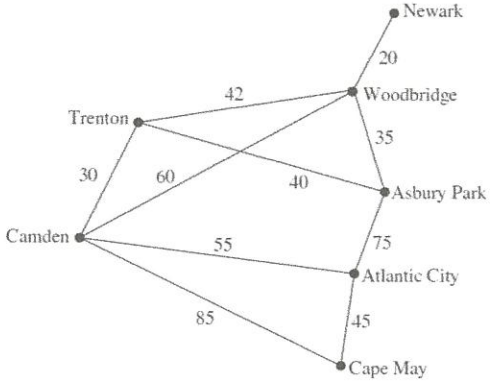

Figure Q5(b)

Find the shortest route between Newark and Camden using these roads.

(10 marks)

- END OF QUESTIONS –

TERBUKA